CAS Heimdall

P&IDs and instrument loop diagrams

Closed aquaculture pen PEN01 – project 15169, ScaleAQ

Purpose. One set of process and instrument drawings for the CAS pen, split at system level, with off-page connectors between the systems, typical instrument loop diagrams, and an I/O list checked line by line against the cabinet drawings and the PLC project.

Drawing practice. ANSI/ISA-5.1 instrument identification and symbols, and ANSI/ISA-5.4 loop diagrams, as presented in T. R. Kuphaldt, *Lessons In Industrial Instrumentation*, ch. 7, and N. Sands, I. Verhappen (eds.), *A Guide to the Automation Body of Knowledge*, 3rd ed., ISA 2018, ch. 1. Symbols and letters: sheet CAS-PID-001.

Sources.

Functional description	P-15169-CAS-FD-01 rev B (13.05.26)
Alarm list	P-15169-CAS-FD-02 rev A
Framo control spec	0000_CAS_SPEC_CTRLSYS rev B
Oxygen cabinet	Linde SOLVOX DC prop. high flow 4 ch (flowsheet)
Hydraulics	Servi S6377070 / 2939238 rev A1 (18.06.26)
Cabinet drawings	15169-792.001 rev 1 (PLC/RIO1), 792.002 rev 0 (RIO2/3), 874.001 (Framo), 874.004 (drive), 875.001/002 (ATS, distribution)
Cable list	15169-860-001 rev 1
PLC project	scaleaq/heimdall-plc, branch inletControl (CODESYS CAS-Heimdall, PFC200_Sim)
Reference only	CAS prelim P&ID rev B (sketch)

Status. Revision A is drawn from the documents above. Points where the sources disagree, or where something is missing, are marked in red on the sheets and listed on CAS-PID-950. How to re-check the set against its sources: checklist.md next to the .tex file.

Drawing index

Drawing	Title	Type
CAS-PID-000	Cover sheet and drawing index	–
CAS-PID-001	Legend: symbols, letters, lines	Legend
CAS-PID-100	Water inlet and Framo inlet pumps (typical inlet 1–3)	P&ID
CAS-PID-200	Oxygen supply and distribution (Linde cabinet)	P&ID
CAS-PID-300	Inlet-branch oxygenation (typical line 1–3)	P&ID
CAS-PID-400	Pen instruments, level and oxygen control	P&ID
CAS-PID-500	Outlet dampers and hydraulic power unit	P&ID
CAS-PID-600	Utilities: bilge, power, cabinet status	P&ID
CAS-LAY-700	Control system network and I/O nodes	Architecture
CAS-LD-801	Loop LT01: level inside pen, 4–20 mA	Loop
CAS-LD-802	Loop PIC41: oxy pump pressure, ATV650	Loop
CAS-LD-803	Loop FIC41: oxygen flow and valve OV41	Loop
CAS-LD-804	Loop OT01: Orbit 864 over RS-485	Loop
CAS-LD-805	Loop SC31: Framo inlet pump over OPC UA	Loop
CAS-LD-806	Loop XV61/FT61/PT61: outlet cylinder 1	Loop
CAS-LD-807	Loop LS711: bilge level switches inlet 1	Loop
CAS-PID-900	I/O list 1/2: CommonRIO, drawing vs PLC	List
CAS-PID-901	I/O list 2/2: InletRIO 1–3, drawing vs PLC	List
CAS-PID-950	Open points	List

Numbering. n = inlet / oxygen line 1, 2, 3. PLC index i = n − 1.
Typical sheets are drawn for n = 1; inlets 2 and 3 are identical with the loop numbers and node names in the table on each sheet.

Title: **Cover sheet and drawing index**	Drawing no.: **CAS-PID-000**	
System: All systems	Project: 15169 CAS Heimdall ScaleAQ	Rev. / date: A / 2026-10-16

Instrument symbols (ISA-5.1)

Outline = kind of device, mid-line = where it is accessible [Kuphaldt p.545].

	Field mounted	Main control (PLC HMI)	Behind panel (not on HMI)	Local panel
Discrete instrument	LT 01	LI 01	LY 01	HS 31
Shared display / control (PLC)	OT 01	LIC 01	–	–
PLC logic function		UY 41	I 1	
Computer function / package PLC		SC 31		

On these sheets: every PLC function with a faceplate on the operator HMI (indicators, controllers, alarms) is a shared-control bubble with a solid mid-line. Calculations the operator does not see are drawn behind the panel (dashed mid-line). Functions inside a supplier's package PLC (Framo Beckhoff controller, ATV650 drive logic) are drawn as computer functions (hexagon). Interlocks are diamonds numbered I1, I2, ... and described in the note list of the sheet.

Identification letters used (ISA-5.1)

Letter	First letter (measured / initiating)	Succeeding letter
A	Analysis (CO_2)	Alarm
C	–	Control
D	Differential (modifier)	–
F	Flow	–
H	Hand	High
I	Current	Indicate
K	–	Control station
L	Level	Low, light
O	**User-defined: dissolved oxygen**	–
P	Pressure	–
Q	Quantity	Integrate / totalise
S	Speed	Switch
T	Temperature	Transmit
U	Multivariable	Multifunction
V	–	Valve, damper
X	Unclassified	–
Y	Event, state	Relay, compute
Z	Position	Driver, actuator

Project conventions

Prefix	All instrument tags are `PEN01-`*tag* (pen 1); the prefix is left out of the bubbles.
O (user-defined)	Dissolved oxygen, as in the FD: OT = O_2 transmitter, OIC = O_2 controller.
OV41–43	Oxygen flow control valves. Project tag (FD, cabinet drawings); in strict ISA terms these are the FV of loops FIC41–43.
XV	On/off valve (solenoid).
Loop numbers	0x pen level; 1x pen analysers; 3x inlet n (Framo); 4x oxygen line n; 6x outlet; 7x bilge inlet n, chamber m (LS7nm).
Equipment	PM-3n Framo inlet pump, PU4n oxy pump, MX-4n O_2 injector, TV-3n vacuum chamber, EJ-3n ejector, XF-3n intake.
n	Inlet / line 1–3. Typical sheets show n = 1.

Lines

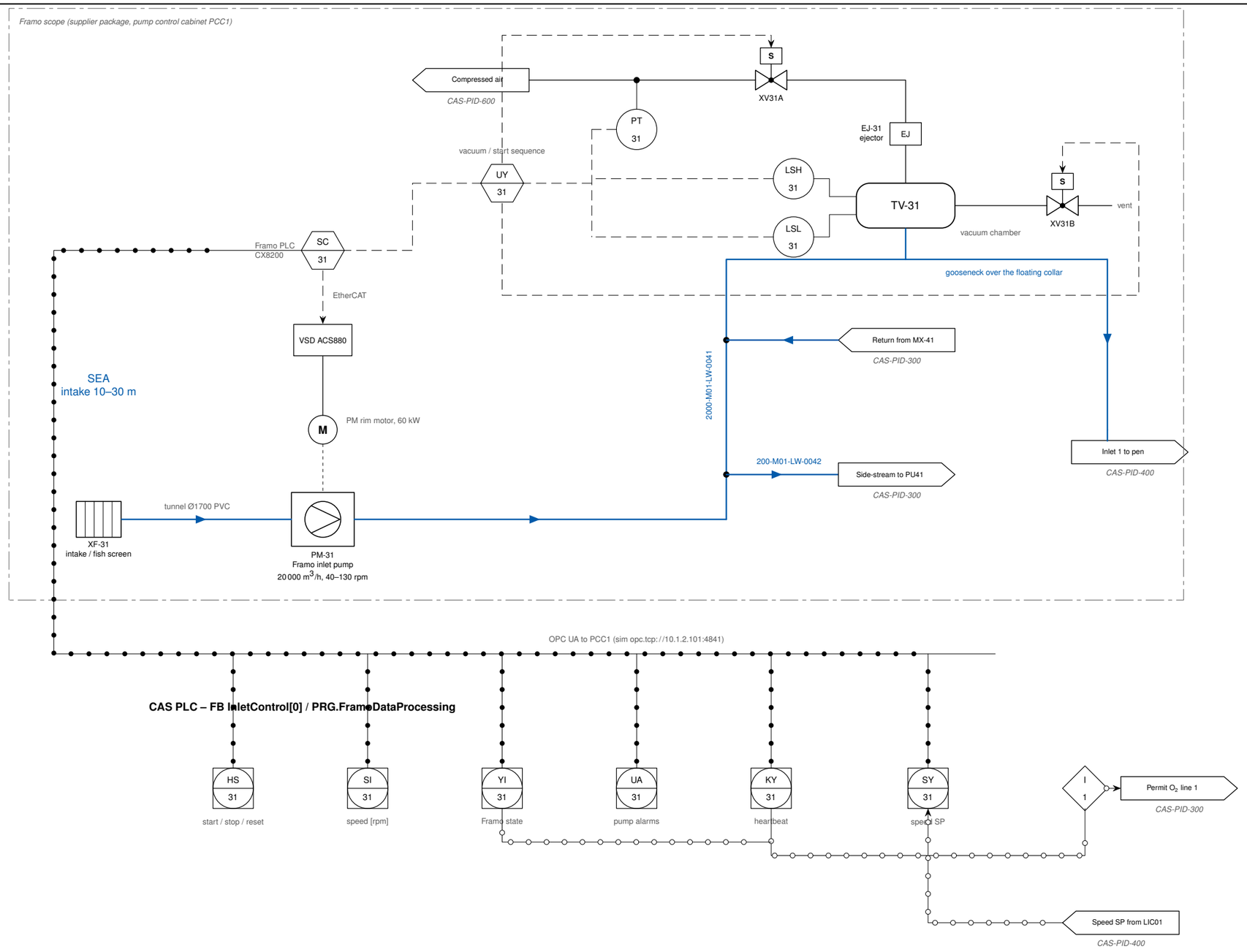
- Process line, sea water
- Process line, oxygen gas
- Process line, hydraulic oil
- Process connection to an instrument
- Electric signal (4–20 mA, 24 V DC, relay contact)
- Fieldbus network: Modbus TCP, Modbus RTU (RS-485)
- Software link inside the CAS PLC
- Data link to an independent system: OPC UA (Framo)
- Mechanical link
- Scope / package boundary (supplier)

ISA-5.1-2009 line types [Kuphaldt p.544]. ISA does not name protocols; the protocol is written next to each network line. Line number format: `size-area-service-number`, e.g. `2000-M01-LW-0041` (DN 2000, pen M01, sea water, line 0041); services LW sea water, LO oxygen, LH hydraulic oil.

Equipment

- Centrifugal pump
- Framo inlet pump (rim driven)
- M — Electric motor
- VSD — Variable speed drive
- Control valve (globe), diaphragm
- S — Valve, solenoid actuator
- Manual valve
- Check valve (flow left to right)
- In-line flow element
- Hydraulic cylinder
- Stream → *continues on* — Off-page connector, outgoing
- Stream ← *comes from* — incoming
- open point — Sources disagree or value not confirmed: see CAS-PID-950

Title: **Legend: symbols, identification letters and lines**		Drawing no.: **CAS-PID-001**
System: All systems	Project: 15169 CAS Heimdall ScaleAQ	Rev. / date: A / 2026-10-16



Notes

1. Framo inlet pump with local controller PCC1 (Beckhoff CX8200; ABB ACS880 VSD over EtherCAT). The CAS PLC writes Start, Stop, Reset, SpeedSetpoint and HeartbeatToggle and reads states, alarms and drive values over OPC UA (PRG.FramoDataProcessing; Framo spec 0000_CAS_SPEC_CTRLSYS rev B).

2. Vacuum system: the Framo PLC opens XV31A to run ejector EJ-31 until LSL31, and on stop opens XV31B to break the vacuum so the pen does not drain back (FD 3.2.1.1). LSH31 overfill: alarm in CAS after 5 s. V001/V002, LSL01/LSH02 and air pressure PT31 are read over OPC UA.

3. SY31 in CAS: speed setpoint = LIC01 output in REMOTE (all pumps get the same speed); actual speed while starting, frozen in HOLD. In START and HOLD Framo runs its own preset speed.

4. KY31: the PLC sets Framo_1.HeartbeatToggle 1 s after Framo cleared it; no clear for 5 s = HeartbeatTimeout: pump shown UNAVAILABLE, no start sent, I1 drops.

5. Cabinet drawing 874.001 shows a Framo SX1000; the FD and the folder have SR2000.

Interlocks

I1. Oxygenation permit line 1 = PM-31 in REMOTE AND heartbeat OK (InletPump.Status.EnableOxyControl → OxyControl.InletPermit, CAS-PID-300).

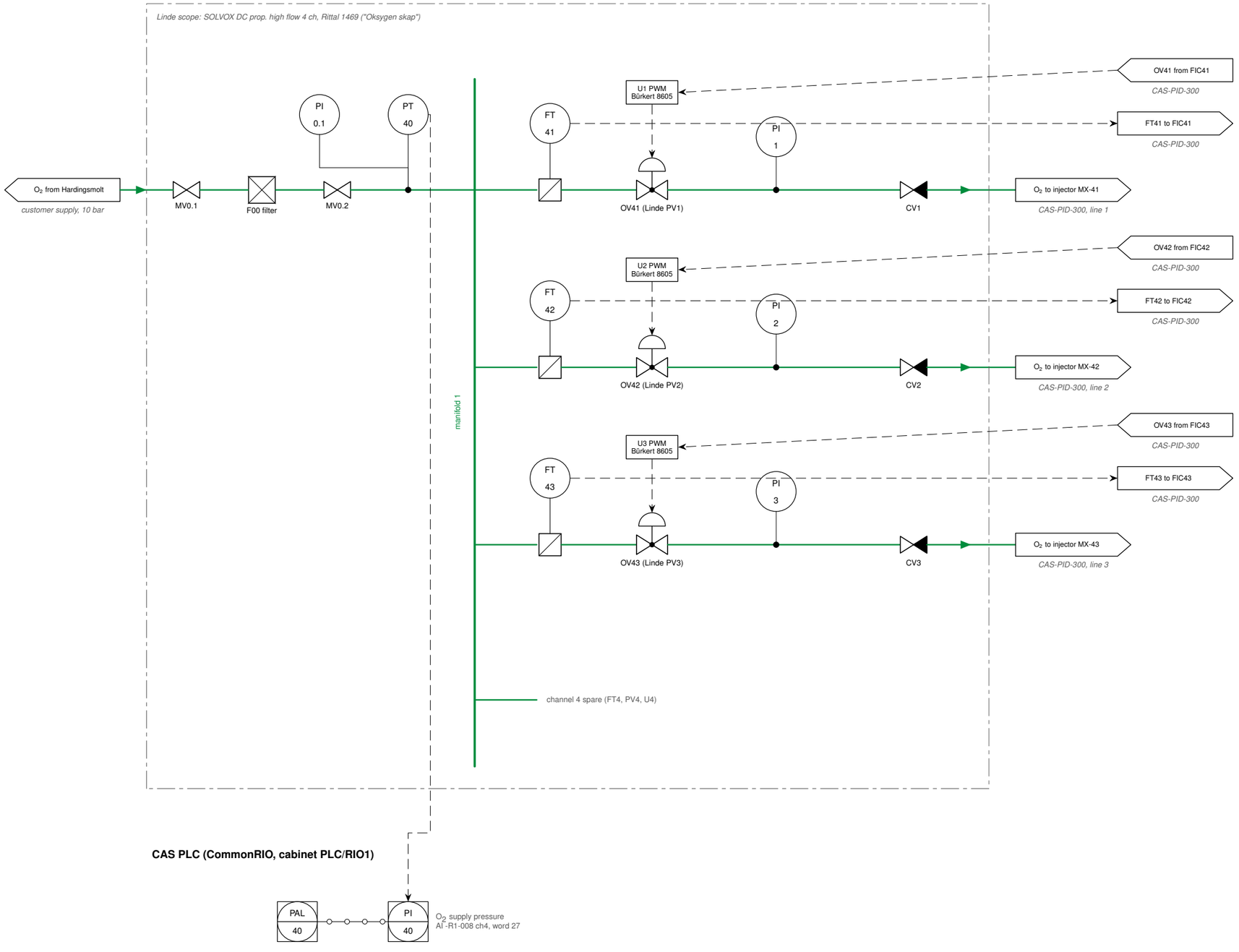
I2. FD 3.2.6: 2oo3 LS71mHH in inlet box 1 stops PM-31. Not in the PLC (CAS-PID-600).

I3. PLC function flow diagram: group start only with the outlet open. Not in the PLC.

Typical

n	1	2	3
Pump, intake	PM-31, XF-31	PM-32, XF-32	PM-33, XF-33
Vacuum system	TV-31, XV31A/B	TV-32, XV32A/B	TV-33, XV33A/B
Switches	LSH31, LSL31	LSH32, LSL32	LSH33, LSL33
Main line	LW-0041	LW-0043	LW-0045
Framo cabinet	PCC1	PCC2	PCC3
PLC	Inlet[0]	Inlet[1]	Inlet[2]
OPC UA (sim)	:4841	:4842	:4843

Title: **Water inlet and Framo inlet pump (typical inlet 1–3)**	Drawing no.: **CAS-PID-100**	
System: Inlet system, drawn for inlet 1	Project: 15169 CAS Heimdall ScaleAQ	Rev. / date: A / 2026-10-16



Notes

1. Linde SOLVOX dosing cabinet per Linde flowsheet (SOLVOX_DC_Prop_high_flow 4Ch_with mass flow, sheet 1): customer valve MV0.1, filter F00, MV0.2, manifold with PI 0.1, and per channel a mass flow transmitter FT, control valve PV (Bürkert 2865 proportional valve, opening set by the CAS PLC with 4–20 mA through PWM board U, Bürkert 8605), PI and check valve CV. Channels 1–3 are used; Linde PV1–3 / FT1–3 = OV41–43 / FT41–43 in CAS.

2. FT41–43: 4–20 mA, 2-wire, on CommonRIO 750-455 -R1-008 ch1–3 (PLC words 24–26), range 0–150 kg/h in the PLC (ConfigValues OxygenFlowConfig). FD table 14 gives the FIC scaling as 0–20 (unit unclear); confirm the transmitter range.

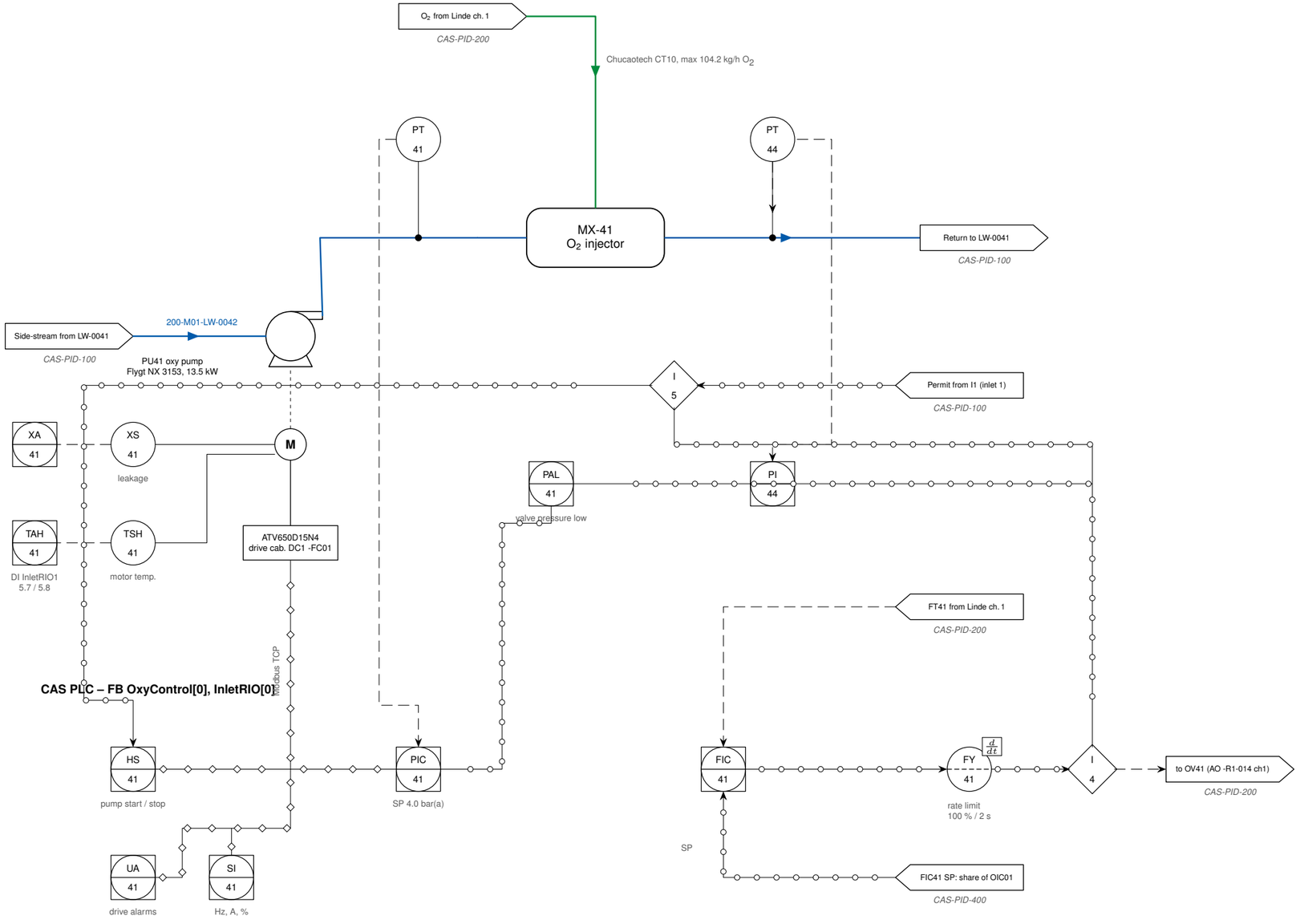
3. OV41–43: 4–20 mA from CommonRIO 750-563 -R1-014 ch1/ch2 and -R1-015 ch1 (PLC out words 20–22), 0–100 % = 0–65535. Fail position: closed (NC valve, no signal = closed).

4. PT40 (O_2 supply pressure, interface to Hardingsmolt): CommonRIO -R1-008 ch4, word 27. The PLC names this input Pen.Input.OxygenInjectionPressure, has no range (DUT default 0–10) and no low alarm; FD/alarm list want PAL40 "oxygen not available".

5. Cable list / drawing 792.001 sheet 14: the FT41 field symbol and cable are labelled FT61 (PLC-PEN01-FT61-KX01 is used twice).

6. PLC function flow diagram has separate "MainOxyValve 1–3" open/close valves; none exist in the Linde flowsheet, the FD or the PLC.

Title: **Oxygen supply and distribution**	Drawing no.: **CAS-PID-200**	
System: Oxygen system: Linde SOLVOX dosing cabinet	Project: 15169 CAS Heimdall ScaleAQ	Rev. / date: A / 2026-10-16



Notes

1. Oxy pump PU41 (Flygt NX 3153 LT, 13.5 kW, 1455 rpm) on a Schneider ATV650D15N4 (drive cabinet DC1). The CAS PLC talks to the drive on its embedded Ethernet port, Modbus TCP: writes CMD 8501 (DRIVECOM) and LFR 8502 (0.1 Hz), reads ETA 3201, RFR 3202, LCR 3204 (0.01 A), THD 3209, OPR 3211 every 100 ms and LFT 7121 every 1 s (FB.RIO.InletRIO, FB.OxySystem.OxyPumpATV650).

2. PIC41: PV PT41 (pressure after the pump, InletRIO1 750-455 ch3 = word 2, 0–10 bar), SP 4.0 bar(a), output pump speed 20–50 Hz. Manual / preset 40 Hz while starting, in pump manual or with bad PV.

3. FIC41: PV FT41 (CAS-PID-200), SP = line 1 share of OIC01's total O_2 flow (CAS-PID-400), output valve position 0–100 %, rate limited by FY41 (ValveRampTime 2 s).

4. PT44: pressure after the injector, InletRIO1 ch4 = word 3, 0–10 bar; indication only. FD table 15 wants a low differential-pressure alarm across the injector (PT44–PT41); not in the PLC. FD numbers the transmitters PT41/43/45 (in) and PT42/44/46 (out); the cabinet drawings and cable list use PT41–43 after the pump and PT44–46 after the injector – drawn as the cabinet drawings.

5. XS41 / TSH41: Flygt MiniCAS II leakage / thermal relay in DC1, dry contacts to InletRIO1 DI 5.7 / 5.8. Read but not used by any PLC logic; RIO2/3 have no such inputs on the drawings.

Interlocks

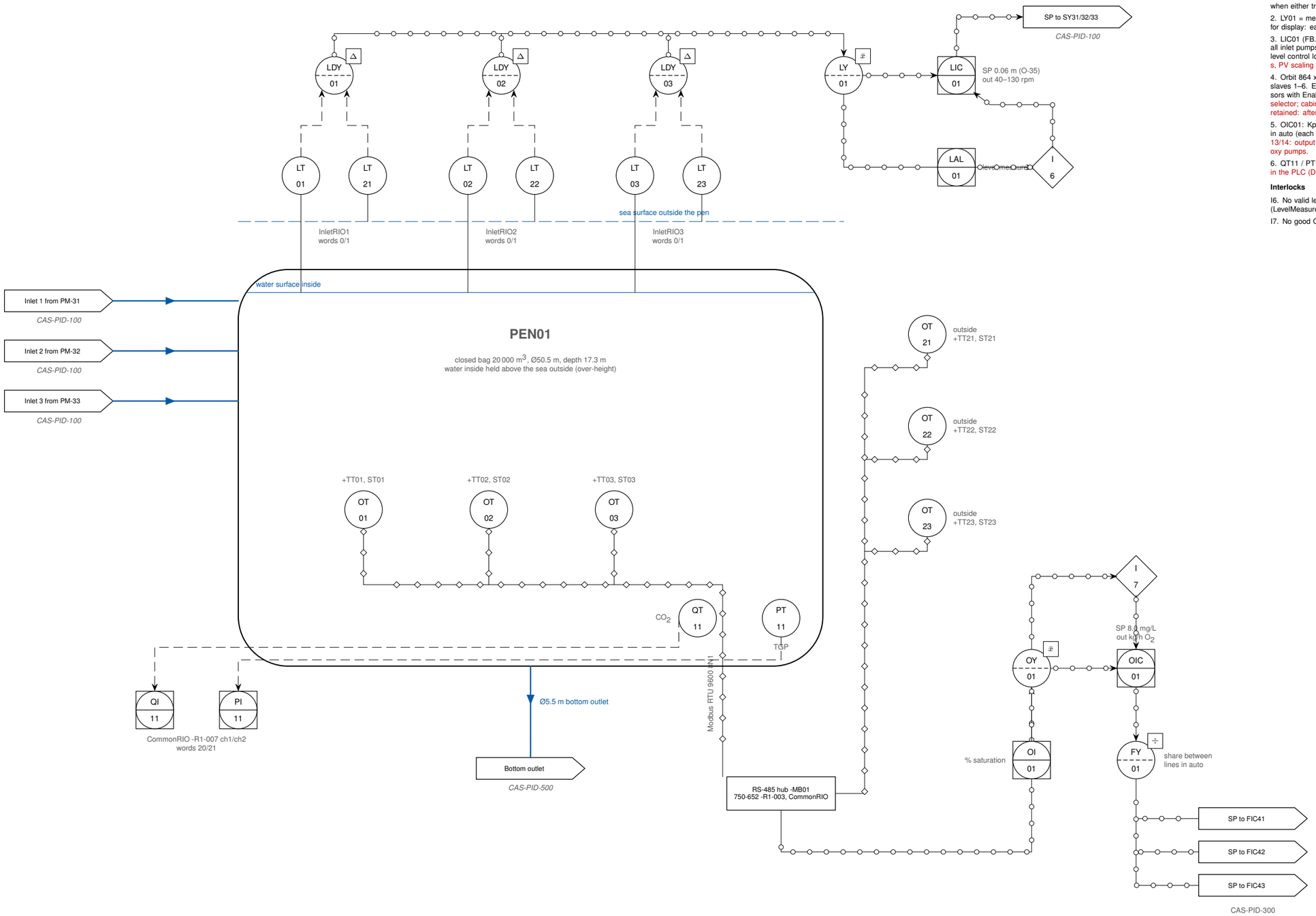
I5. Line start: operator EnableOxyControl AND permit I1 from inlet 1 AND NOT emergency stop → oxy pump starts in auto. EmergencyStop is not wired to any input.

I4. Valve permit: pump running for RunDelay (30 s) after line start AND PT41 good AND PT41 > 3.0 bar (ValveOpenPressure); otherwise OV41 is forced closed (4 mA), also in valve manual. Oxygenation active without the permit raises PAL41 (ValvePressureLow).

Typical

n	1	2	3
Pump, drive	PU41, DC1	PU42, DC2	PU43, DC3
Injector	MX-41	MX-42	MX-43
Pressure after pump / injector	PT41 / PT44	PT42 / PT45	PT43 / PT46
Controllers	PIC41, FIC41	PIC42, FIC42	PIC43, FIC43
Flow, valve	FT41, OV41	FT42, OV42	FT43, OV43
Branch line	LW-0042	LW-0044	LW-0046
RIO node	InletRIO1	InletRIO2	InletRIO3
Drive (sim)	:5131	:5132	:5133
PLC	OxySystem[0]	OxySystem[1]	OxySystem[2]

Title: **Inlet-branch oxygenation (typical line 1–3)**	Drawing no.: **CAS-PID-300**	
System: Oxygenation, drawn for line 1	Project: 15169 CAS Heimdall ScaleAQ	Rev. / date: A / 2026-10-16



Notes

1. Level: inside LT01–03 (VEGAPULS radar on the gooseneck), outside LT21–23 (submersible pressure probe), one pair per inlet on InletRIO n words 0/1, 0–2 m, 2-wire 4–20 mA. LDY0n = over-height of pair n = LT0n − LT2n from the unfiltered values, then a Butterworth low-pass (FilterFc 0.05 Hz); a pair drops out when either transmitter is bad or disabled (FB.Inlet.InletSensors).

2. LY01 = mean over-height of the valid pairs = LIC01 PV (FB.Pen.PenControl). Inside / outside means for display: each transmitter filtered on its own, then averaged.

3. LIC01 (FB.Inlet.InletSupervisor): Kp 450 rpm/m, Ti 60 s, out 40–130 rpm, common speed setpoint of all inlet pumps in REMOTE. Tracks the actual pump speed while pumps start, freezes when a pump in level control loses its heartbeat (HOLD). FD table 8: P scheduled on the number of pumps (0.5/1/2), I 600 s, PV scaling 0–20 cm, deviation and L/H alarms; the PLC has one gain set and no L/H alarms.

4. Orbit 864 x6 on RS-485 through the 750-652 behind CommonRIO and the Waveshare hub -MB01, slaves 1–6. Each gives O_2 (mg/L, % sat), temperature, salinity and depth. OY01 = mean O_2 of the sensors with EnableForControl = OIC01 PV. FD: inside OT01–03, outside OT21–23, PV normally OT03 with a selector; cabinet drawing: OT01–OT06 on hub ports 1–6. EnableForControl defaults to FALSE and is not retained: after a restart OIC01 has no PV until set online.

5. OIC01: Kp 50, Ti 1800 s, SP 8.0 mg/L; output = total O_2 flow [kg/h], split by FY01 between the lines in auto (each up to FlowMax 104.2 kg/h; lines in manual counted with their measured flow). FD table 13/14: output 0–100 % to OV41–43 directly or as cascade to FIC41–43, P scheduled on the number of oxy pumps.

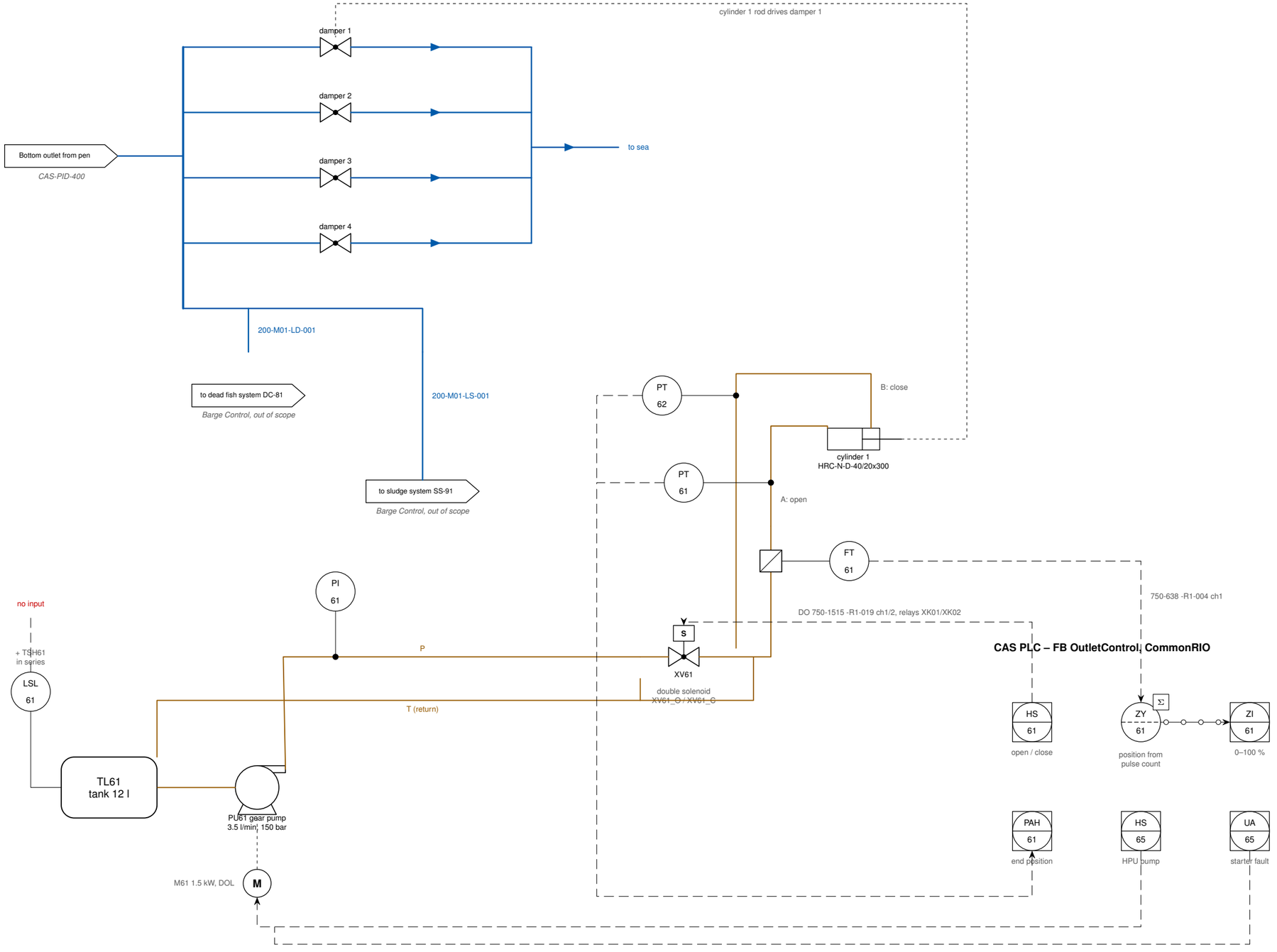
6. QT11 / PT11: PSI Solu-Blu CO_2 / total dissolved gas pressure, 2-wire 4–20 mA. No engineering range in the PLC (DUT default 0–10). Drawing 792.001 shows QT11 with an "O2 transmitter" symbol.

Interlocks

I6. No valid level pair (LevelControlOK FALSE): LIC01 to manual at SafeOutput 60 rpm, alarm LAL01 (LevelMeasurementLost); back to auto bumplessly.

I7. No good O_2 sensor for control (OxyControlOK FALSE): OIC01 to manual, output frozen.

Title: **Pen instruments, level control LIC01 and oxygen control OIC01**	Drawing no.: **CAS-PID-400**	
System: Pen PEN01	Project: 15169 CAS Heimdall ScaleAQ	Rev. / date: A / 2026-10-16



Notes

1. Outlet: Ø5.5 m bottom outlet with four dampers, each moved by a hydraulic cylinder from the Servi HPU (hydraulic schematic S6377070 / 2939238 rev A1). Fail position wanted: close (FD 3.2.3, HOLD). Dead fish and sludge pumps are controlled by Barge Control, not by the CAS PLC.

2. Cylinder n: XV6n double-solenoid valve (Hawe PSL), XV6n_O on 750-1515 -R1-019 ch 2n−1, XV6n_C on ch 2n, via interposing relays XK; oil-flow meter FT6n (HySense QG100, ≈≈5200 pulses/l, 2080 pulses full stroke) on 750-638 counters -R1-004/005; PT(2n−1) open side and PT(2n) close side on 750-455 -R1-009/010, end position by high pressure (FB.Outlet.OutletControl).

3. HPU pump PU61/M61: direct-on-line starter, PLC start = DO word 24.9, reset = 24.10, starter fault = DI word 40.0. Drawing 792.001 R1: these channels are spare (-R1-020 ch2/3) and DI -R1-017 is all spare; with six 750-455 modules on CommonRIO the DI word is 44, not 40.

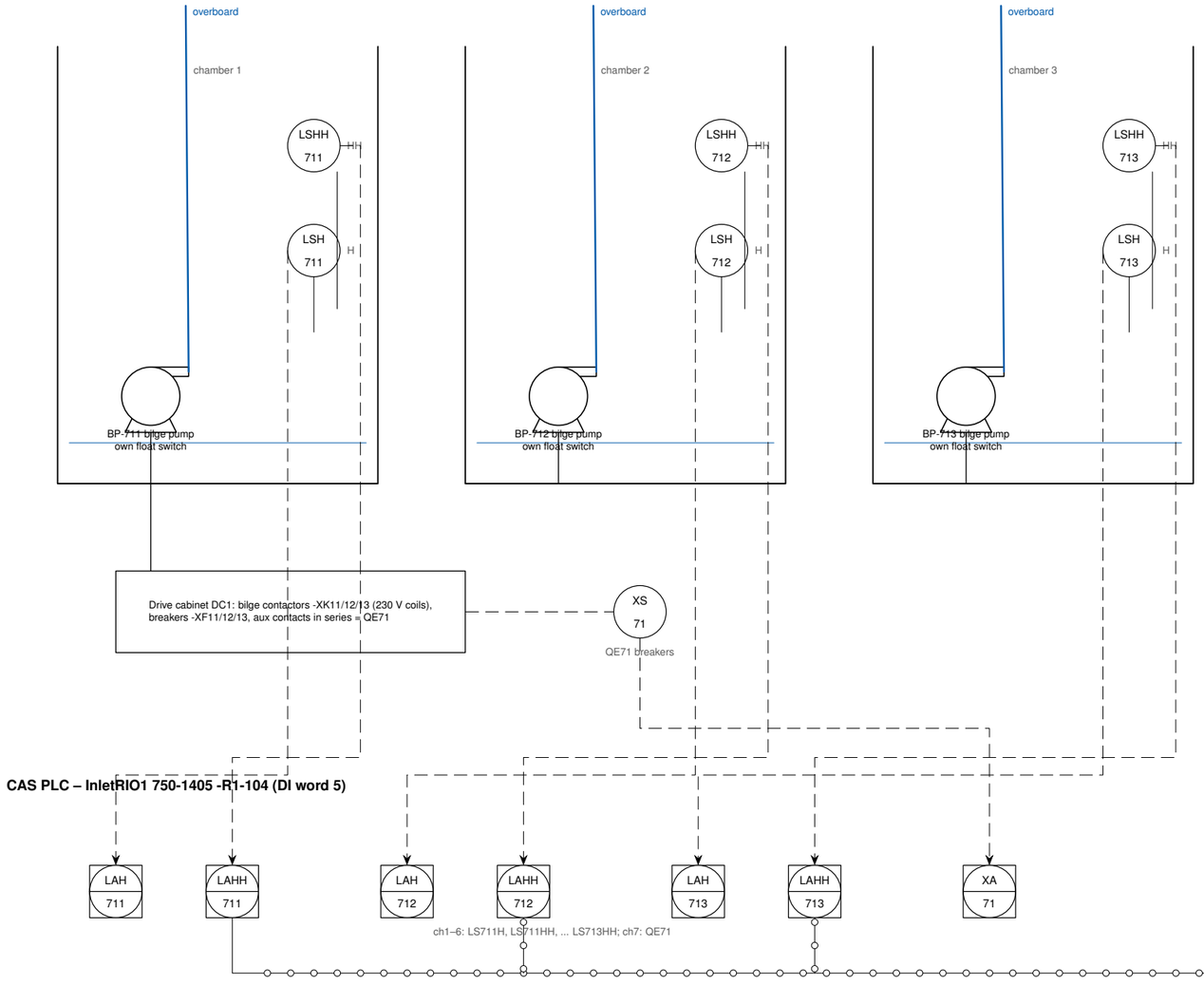
4. LSL61 + TSH61 (tank level / temperature, in series): FD wants a DI; no input on the drawings and Outlet.Input.HydraulicOilTankFault is not mapped in the PLC. XV65 shut-off valve was removed from drawing R1, but its cable is still in the cable list; cable list also has PT69–PT612.

Typical

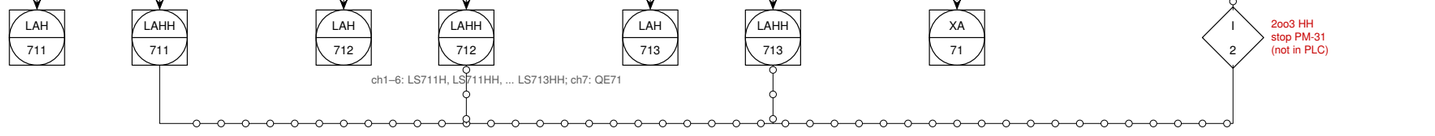
Cylinder	1	2	3	4
Valve	XV61	XV62	XV63	XV64
DO ch (R1-019)	1 / 2	3 / 4	5 / 6	7 / 8
Flow meter	FT61	FT62	FT63	FT64
Counter	R1-004 ch1	R1-004 ch2	R1-005 ch1	R1-005 ch2
PT open / close	PT61 / 62	PT63 / 64	PT65 / 66	PT67 / 68
AI module	R1-009	R1-009	R1-010	R1-010
PLC	Cylinder[0]	Cylinder[1]	Cylinder[2]	Cylinder[3]

Title: **Outlet dampers and hydraulic power unit**		Drawing no.: **CAS-PID-500**
System: Outlet system	Project: 15169 CAS Heimdall ScaleAQ	Rev. / date: A / 2026-10-16

Inlet box 1 (pump module 1): three chambers

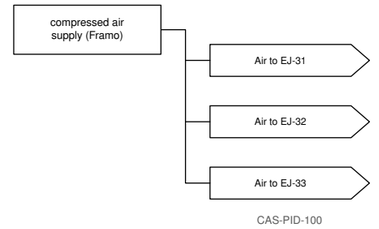


CAS PLC – InletRIO1 750-1405 -R1-104 (DI word 5)



Bilge pump contactors: DC sheet 05 expects RIO relays -XK1/2/3 (230 V); no RIO drawing has them, the PLC has no outputs for them.

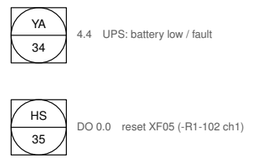
Compressed air



CAS-PID-100

Source not on any drawing
(prelim P&ID: "From compressed air <dwg no.>").

Cabinet power status (InletRIO n, -R1-101, DI word 4)



RIO2/RIO3: 4.3 and 4.4 are wired to +24 V (no UPS in those cabinets) and always read TRUE. The PLC maps them as BackupPowerOn / BatteryFault. Only FieldPower24VTripped (4.0) is used by the PLC.

Notes

1. Each inlet box has three chambers with a self-controlled bilge pump and two level switches (VEGAPOINT 11): LS7nmH water detected (alarm), LS7nmHH high-high (FD 3.2.6, table 16). n = inlet box, m = chamber.

2. Switches on InletRIO n 750-1405 -R1-104 ch1–6, bilge breakers QE7n on ch7, 2-wire 24 V (drawings 792.001 sh.29, 792.002 sh.12). PLC: InletRIO words 5.0–5.6). The PLC reads them but raises no alarm and has no 2oo3 trip.

3. Power: shore / barge supply via the ATS cabinet (875.001, no signals to the PLC). UPS (ADEL CBI245A, 2 x 12 V 26 Ah) only in the PLC/RIO1 cabinet.

Typical

n	1	2	3
Switches	LS711..713H/HH	LS721..723H/HH	LS731..733H/HH
Breakers	QE71	QE72	QE73
RIO	InletRIO1	InletRIO2	InletRIO3
Drive cabinet	DC1	DC2	DC3

Title	**Utilities: bilge, compressed air, cabinet power status**	Drawing no.	**CAS-PID-600**
System	Utilities, bilge drawn for inlet box 1	Project	15169 CAS Heimdall ScaleAQ
		Rev. / date	A / 2026-10-16

Network cabinet NC: Ethernet switch (not drawn on any sheet) – Cat6 star, cable list NC-xxx-KY01

Only the Cat6 (NC-PLC/RIO1-KY01) to the PLC/RIO1 cabinet for PFC200, R1-001 and R1-100: internal link not drawn.

Modbus TCP

OPC UA

PLC/RIO1 cabinet =F1+AC (15169-792.001 rev 1), UPS ADEL CBI245A

-R1-000 PFC200 750-8212 CAS PLC

-R1-001 750-362 coupler | -R1-002 750-602 | -R1-003 750-652 | -R1-004 750-638 | -R1-005 750-638 | -R1-006 750-602 | -R1-007 750-455 | -R1-008 750-455 | -R1-009 750-455 | -R1-010 750-455 | -R1-011 750-455 | -R1-012 750-455 | -R1-013 750-602 | -R1-014 750-563 | -R1-015 750-563 | -R1-016 750-602 | -R1-017 750-1405 | -R1-018 750-602 | -R1-019 750-1515 | -R1-020 750-1515

CommonRIO (PLC FB.RIO.CommonRIO, sim 10.1.2.48:502): 750-652 Orbit RS-485 | 2x 750-638 cylinder counters | 6x 750-455 AI | 2x 750-563 AO | 750-1405 DI | 2x 750-1515 DO

Drawing: six 750-455 (-R1-007..012, two spare). PLC: five (words 20–39), DI at word 40 – with six the DI is word 44.

-R1-100 750-362 coupler | -R1-101 750-1405 | -R1-102 750-1515 | -R1-103 750-602 | -R1-106 750-455 | -R1-105 750-602 | -R1-104 750-1405

InletRIO1 (PLC InletRIO[0], sim 10.1.2.101:5121)
AI words 0–3: LT01, LT21, PT41, PT44 (-R1-106)
DI word 4: cabinet status (-R1-101), word 5: LS711..713H/HH, QE71, PU41 leakage / temperature (-R1-104); DO word 0.0: reset XF05 (-R1-102)

-MB01 RS-485 hub, 8 ports

Modbus RTU 9600 8N1 to Orbit 864 x6 (OT01–06 on ports 1–6), slaves 1–6

RIO2 cabinet (15169-792.002 rev 0), no UPS

-R1-100 750-362/040 | -R1-101 750-1405 | -R1-102 750-1515 | -R1-103 750-602 | -R1-106 750-455 | -R1-105 750-602 | -R1-104 750-1405

InletRIO2 (InletRIO[1], sim :5122): LT02, LT22, PT42, PT45;
LS721..723H/HH, QE72. No PU42 leakage / temperature DIs.

RIO3 cabinet (15169-792.002 rev 0), no UPS

-R1-100 750-362/040 | -R1-101 750-1405 | -R1-102 750-1515 | -R1-103 750-602 | -R1-106 750-455 | -R1-105 750-602 | -R1-104 750-1405

InletRIO3 (InletRIO[2], sim :5123): LT03, LT23, PT43, PT46;
LS731..733H/HH, QE73. No RIO3 drawing; from the cable list.

Drive cabinet DC1 ATV650D15N4 -FC01, PU41

Drive cabinet DC2 ATV650D15N4 -FC01, PU42

Drive cabinet DC3 ATV650D15N4 -FC01, PU43

ATS cabinet
no signals to the PLC

Framo PCC1 CX8200 + ACS880, PM-31

Framo PCC2 CX8200 + ACS880, PM-32

Framo PCC3 CX8200 + ACS880, PM-33

DC n: Modbus TCP on the embedded Ethernet port, polled by InletRIO[i] (2nd SCL master), unit id 0, sim 10.1.2.101:5131–5133.
PCC n: OPC UA server (Framo spec), read / written by PRG.FramoDataProcessing, sim 10.1.2.101:4841–4843.

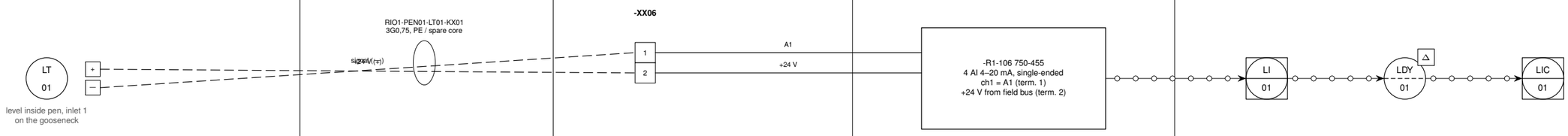
- 750-455 AI 4–20 mA
- 750-563 AO 4–20 mA
- 750-1405 DI / 750-638 counter
- 750-1515 DO
- 750-602 24 V feed
- coupler / CPU / 750-652 serial

Notes

1. The PFC200 has no local I/O. All field I/O is on 750-362 Modbus TCP couplers polled by the PLC program: CommonRIO FC23 read 41 / write 25 words (WagoAppPlcModbus), InletRIO n FC23 read 26 / write 21 words every 100 ms (SCL.sModbusMasterTCP). The 750-652 behind CommonRIO carries Modbus RTU in the coupler's process image (words 0–11).
2. Naming: the drawings' "RIO1" is the inlet-1 coupler -R1-100 (PLC InletRIO[0]); the PLC's CommonRIO is coupler -R1-001 in the same cabinet.
3. IP addresses are not on any drawing. The values shown are the simulator defaults in the PLC (PFC200_Sim GVL ConfigValues).
4. Process image order of a 750-362: complex and analog modules first in slot order, then digital modules packed bit-wise. Word numbers in this set follow the PLC mapping; CAS-PID-900 compares them with the drawings.

Title: **Control system network and I/O nodes**	Drawing no.: **CAS-LAY-700**
System: Control system: PLC, Modbus TCP couplers, drives, Framo	Project: 15169 CAS Heimdall ScaleAQ
	Rev. / date: A / 2026-10-16

Field	Field cable	Cabinet terminals	I/O module	PLC function / operator HMI



cal. 0–2 m
= 4–20 mA

24 V loop supply
750-602 -R1-103 field bus
← XF05 OUT+6 (CBM E8, 2 A, shared)
← XT02 QUINT DC/DC ← XT01 UPS (2 x 12 V 26 Ah)
0 V: module 0 V to -XX0V

PLC: GVLMain.Inlet[0].Input.LevelInsidePen
(AnalogSensor: 0–2 m, FilterFc 0.05 Hz,
quality alarm). Pair with LT21 in LDY01,
mean of pairs = LIC01 PV (CAS-PID-400).

2-wire loop-powered transmitter: + to the +24 V terminal, — to the signal terminal, current returns through the 750-455 input to 0 V. Wire numbers / colours are not on the drawings or in the cable list.

Coupler -R1-100 750-362 (InletRIO1)
Modbus TCP FC23, input word 0
bits 1..0 = diagnostics, value bits 14..4

Tag	Description	Make / model	Range / signal	Location	Source
LT01	Level transmitter, inside pen at inlet 1	VEGA VEGAPULS C22 (radar)	0–2 m / 4–20 mA 2-wire	gooseneck inlet 1	FD 3.2.2; sensor overview; ConfigValues
-XX06 1/2	Field terminals, InletRIO1	Phoenix	–	PLC/RIO1 cabinet	792.001 sh.31
-R1-106 ch1	Analog input	WAGO 750-455	4–20 mA, word 0	PLC/RIO1 cabinet	792.001 sh.31; FB.RIO.InletRIO.InputMapping
LI01, LDY01, LIC01	Indication, over-height pair 1, level controller	CAS PLC (PFC200)	0–2 m	PLC	FB.Inlet.InletSensors, FB.Inlet.InletSupervisor

Typical: LT02/LT03 on InletRIO2/3 (-XX06 1/2), LT21–23 on -XX06 3/4 (ch2, word 1), PT41–43 on 5/6 (ch3, word 2), PT44–46 on 7/8 (ch4, word 3).

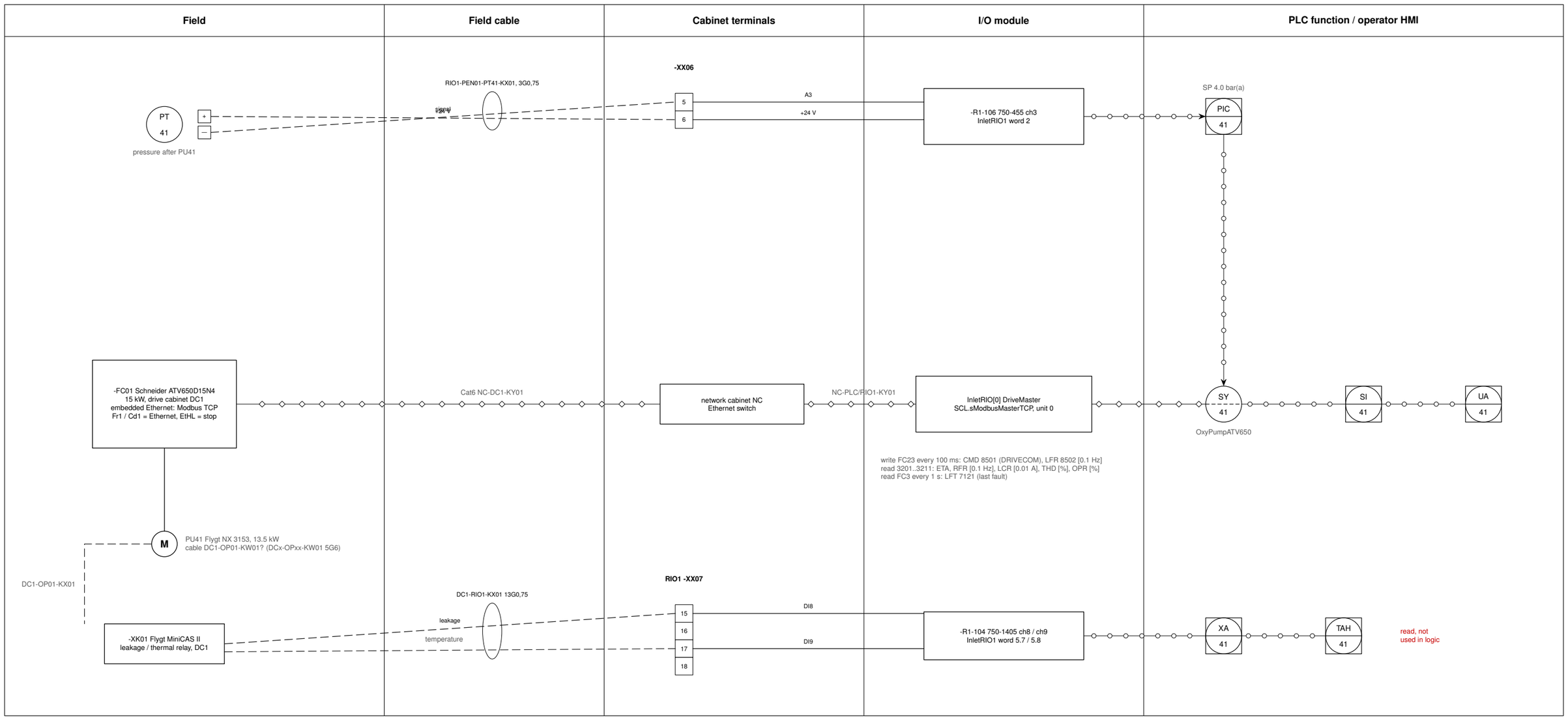
Title
Loop LT01: level inside pen, inlet 1 (typical 2-wire 4–20 mA input)

Drawing no.
CAS-LD-801

System
Loop diagram, typical for LT01–03, LT21–23, PT41–46

Project
15169 CAS Heimdall
ScaleAQ

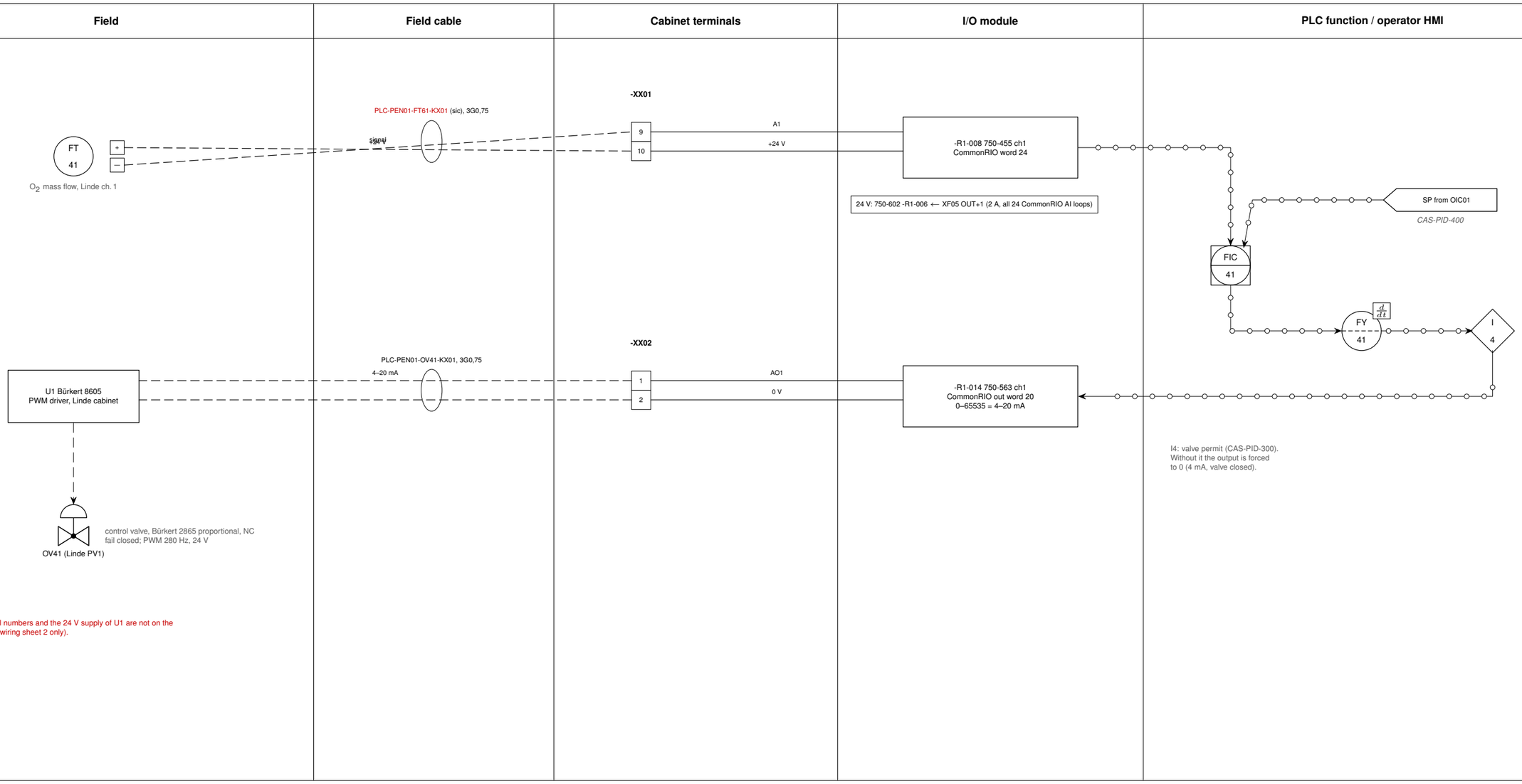
Rev. / date
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Tag	Description	Make / model	Range / signal	Location	Source
PT41	Pressure transmitter, after oxy pump PU41	not given	0–10 bar (PLC) / 4–20 mA 2-wire	branch line 1	792.001 sh.31; ConfigValues
PIC41	Pressure controller, output pump speed 20–50 Hz	CAS PLC	SP 4.0 bar(a)	PLC	FB.OxySystem.OxyControl
-FC01	Variable speed drive PU41	Schneider ATV650D15N4	15 kW, Modbus TCP	drive cabinet DC1	874.004 sh.04; EAV64332; EAV64318
XS41, TSH41	Pump leakage / motor temperature (MiniCAS II)	Flygt MiniCAS II	dry contacts	DC1	874.004 sh.05; 792.001 sh.29

Typical: line 2 / 3 on InletRIO2 / 3 and DC2 / DC3 (sim ports 5132 / 5133). RIO2/3 have no leakage / temperature DIs on the drawings.

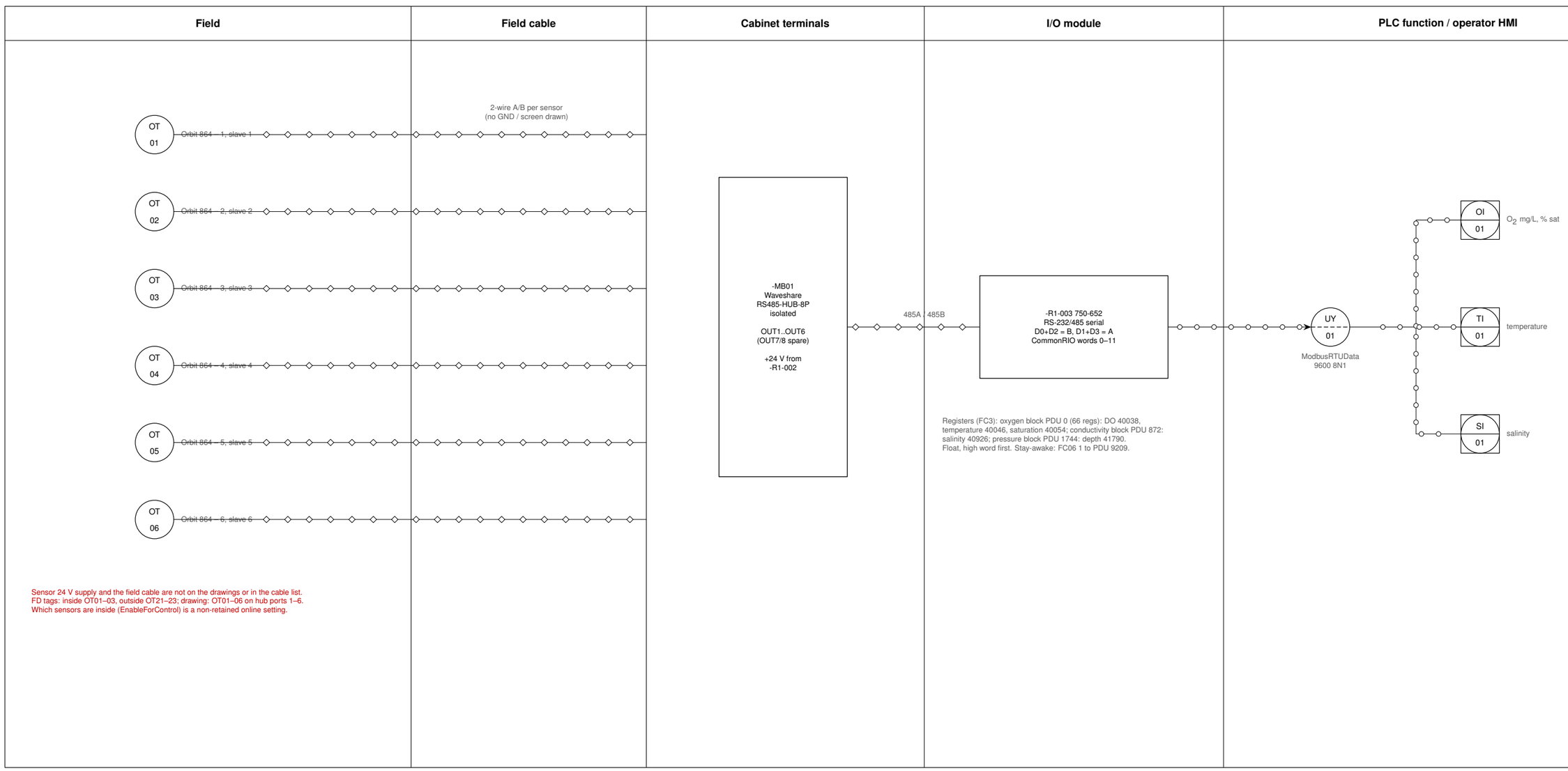
Title: **Loop PIC41: oxy pump pressure line 1, ATV650**	Drawing no.: **CAS-LD-802**	
System: Loop diagram, typical for PIC41–43	Project: 15169 CAS Heimdall ScaleAQ	Rev. / date: A / 2026-10-16



Tag	Description	Make / model	Range / signal	Location	Source
FT41	O_2 mass flow transmitter, Linde channel 1	Linde (SUTO S418 type), to confirm	0–150 kg/h (PLC) / 4–20 mA	Linde cabinet	Linde flowsheet; 792.001 sh.14
FIC41, FY41	Flow controller, rate limit 100 % / 2 s	CAS PLC	0–100 %	PLC	FB.OxySystem.OxyControl
U1, OV41	PWM driver and proportional valve	Bürkert 8605 + 2865	4–20 mA → 0–100 %	Linde cabinet	Linde sheet 2; 792.001 sh.20

Typical: FT42/43 on -XX01 11/12, 13/14 (-R1-008 ch2/3, words 25/26); OV42 on -XX02 3/4 (-R1-014 ch2, word 21), OV43 on 5/6 (-R1-015 ch1, word 22).

Title: **Loop FIC41: oxygen flow line 1 and control valve OV41**	Drawing no.: **CAS-LD-803**	
System: Loop diagram, typical for FIC41–43	Project: 15169 CAS Heimdall ScaleAQ	Rev. / date: A / 2026-10-16



Sensor 24 V supply and the field cable are not on the drawings or in the cable list.
FD tags: inside OT01–03, outside OT21–23; drawing: OT01–06 on hub ports 1–6.
Which sensors are inside (EnableForControl) is a non-retained online setting.

Tag	Description	Make / model	Range / signal	Location	Source
OT01..06	O_2 / temperature / salinity / depth multisensor	In-Situ / Orbit 864 (RDO Core)	DO 0–60 mg/L; Modbus RTU	pen, 3 inside, 3 outside	FD table 17; 792.001 sh.11
-MB01	RS-485 hub, 8 isolated ports	Waveshare RS485-HUB-8P	24 V	PLC/RIO1 cabinet	792.001 sh.11
-R1-003	Serial interface behind CommonRIO	WAGO 750-652	RS-485 9600 8N1	PLC/RIO1 cabinet	792.001 sh.11; PFC200_Sim ModbusRTUData

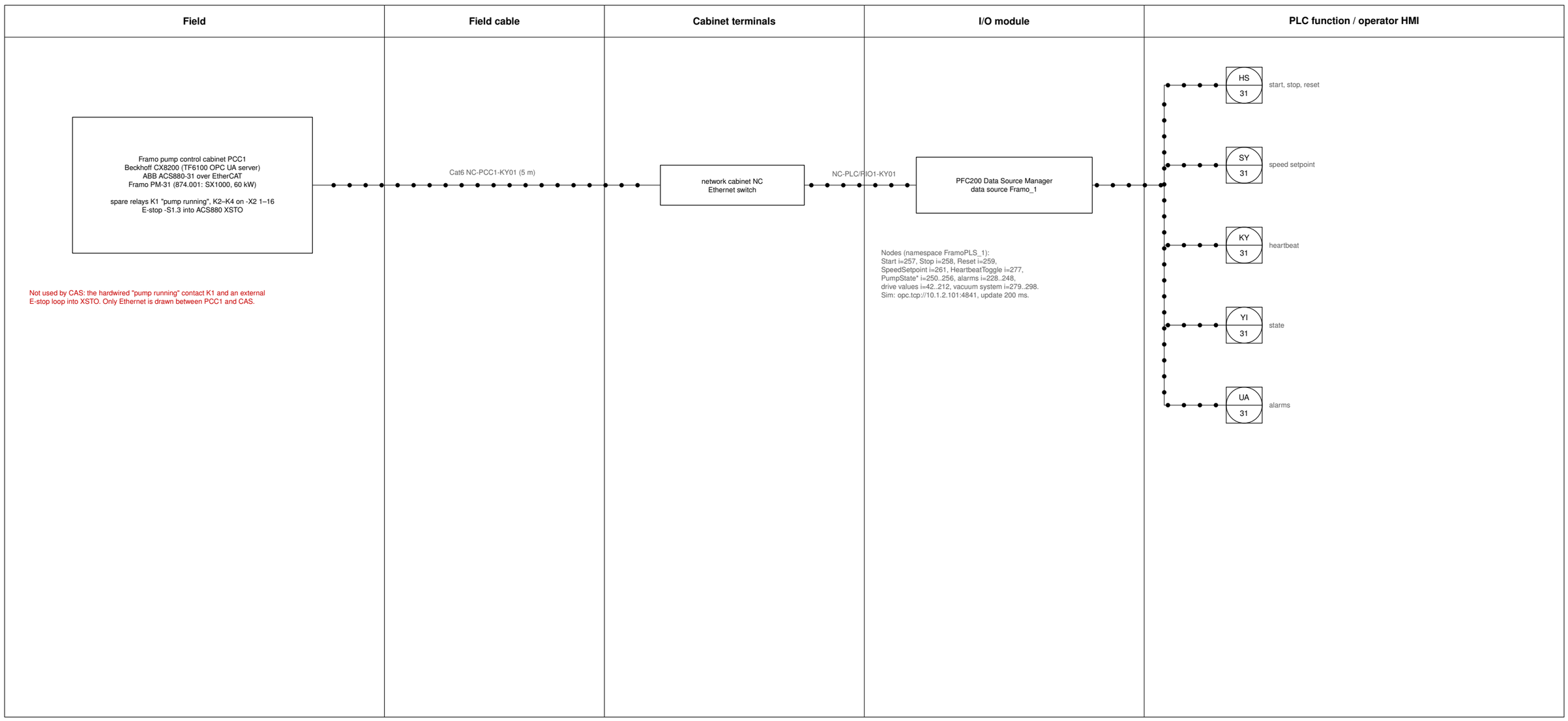
Title
Loop OT01: Orbit 864 multisensor over RS-485 Modbus RTU

Drawing no.
CAS-LD-804

System
Loop diagram, typical for the six Orbit 864 sensors

Project
15169 CAS Heimdall
ScaleAQ

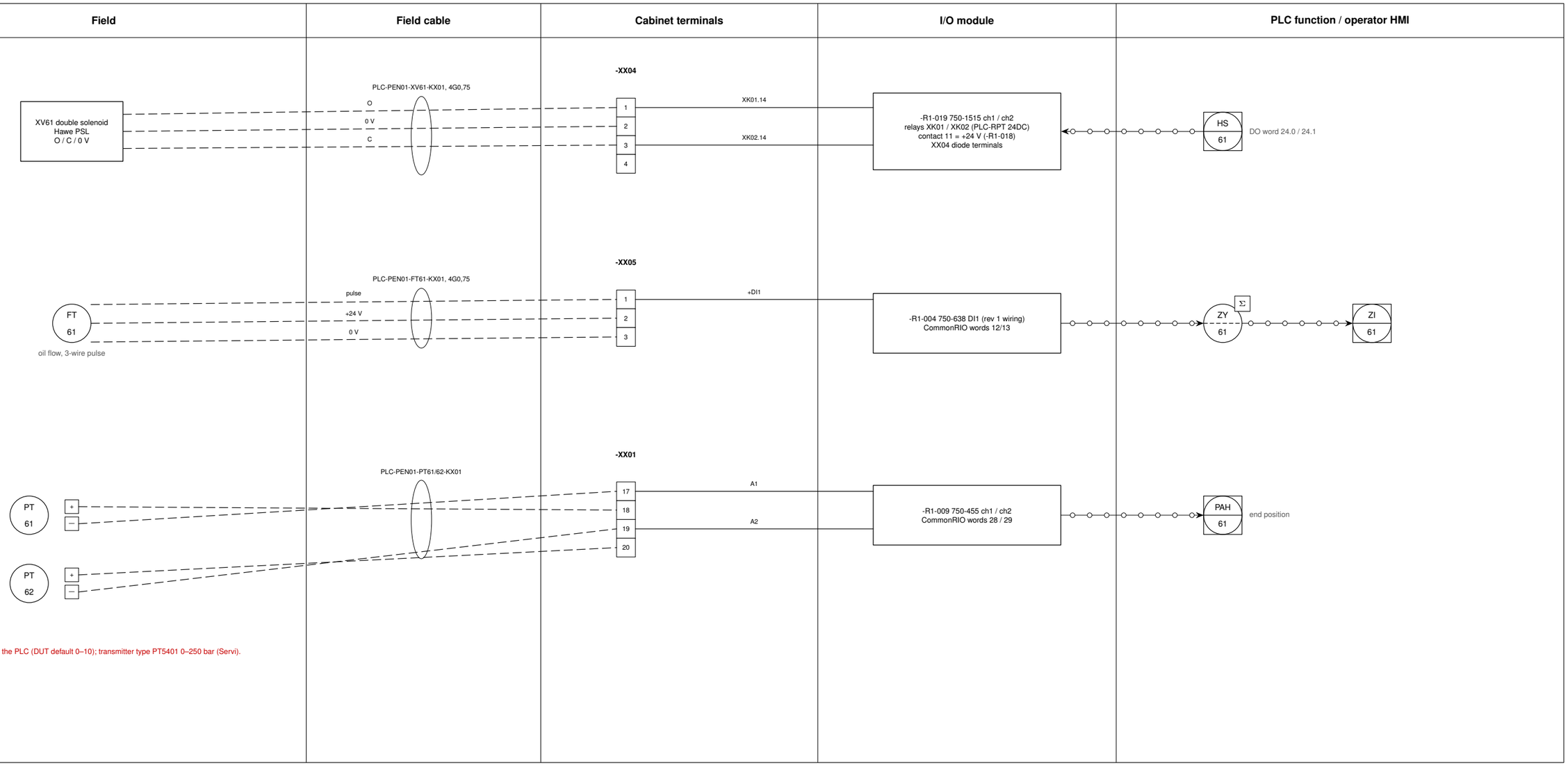
Rev. / date
A / 2026-10-16



Tag	Description	Make / model	Range / signal	Location	Source
PM-31	Framo inlet pump 1 with local controller	Framo SR2000 (FD) / SX1000 (874.001)	40–130 rpm	pump module 1	FD 3.2.1; 874.001
SY31, KY31	Speed setpoint (LIC01 in REMOTE), heartbeat	CAS PLC	rpm; toggle 1 s / timeout 5 s	PLC	FB.Inlet.InletControl; PRG.FramoDataProcessing

Typical: PCC2 / PCC3 (Cat6 NC-PCC2/3-KY01, 60 m), data sources Framo_2 / Framo_3 (sim 4842 / 4843).

Title: **Loop SC31: Framo inlet pump 1 over OPC UA**	Drawing no.: **CAS-LD-805**	
System: Loop diagram, typical for inlet pumps PM-31..33	Project: 15169 CAS Heimdall ScaleAQ	Rev. / date: A / 2026-10-16

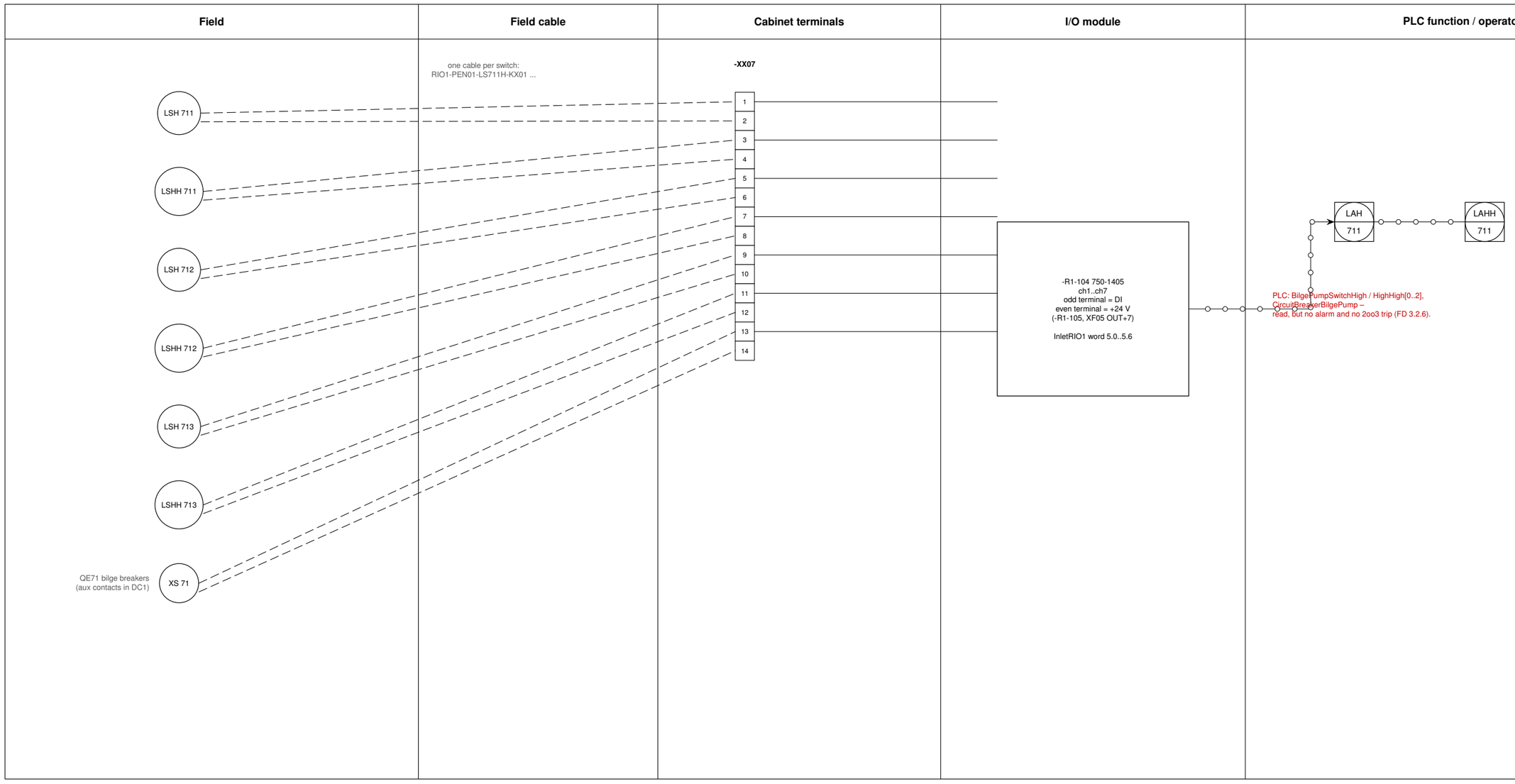


PT61..68 range not in the PLC (DUT default 0–10); transmitter type PT5401 0–250 bar (Servi).

Tag	Description	Make / model	Range / signal	Location	Source
XV61	Directional valve cylinder 1, open / close	Hawe PSL size 2	24 V DC, min 250 mA	hydraulic cabinet	792.001 sh.24; Servi A1
FT61	Oil flow meter cylinder 1 (position by pulse count)	HySense QG100	≈5200 pulses / l	hydraulic cabinet	792.001 sh.10 (rev 1)
PT61 / PT62	Cylinder 1 pressure open / close side	PT5401 (Servi)	0–250 bar / 4–20 mA	hydraulic cabinet	792.001 sh.15

Typical: cylinders 2–4 per the table on CAS-PID-500.

Title: **Loop XV61 / FT61 / PT61-62: outlet cylinder 1**	Drawing no.: **CAS-LD-806**	
System: Loop diagram, typical for cylinders 1–4	Project: 15169 CAS Heimdall ScaleAQ	Rev. / date: A / 2026-10-16



Tag	Description	Make / model	Range / signal	Location	Source
LS711H..LS713HH	Water detected (H) / high-high (HH), chambers 1–3 inlet box 1	VEGA VEGAPOINT 11	24 V switch	inlet box 1	FD table 16; 792.001 sh.29
QE71	Bilge pump breakers, aux contacts in series	–	dry contact	drive cabinet DC1	874.004 sh.05

Typical: inlet boxes 2 / 3 on InletRIO2 / 3 -R1-104 (cables RIO2/RIO3-PEN01-...).

Title: **Loop LS711: bilge level switches, inlet box 1**	Drawing no.: **CAS-LD-807**	
System: Loop diagram, typical for LS7nm H / HH and QE7n	Project: 15169 CAS Heimdall ScaleAQ	Rev. / date: A / 2026-10-16

Module / ch	Drawing: tag, description	Terminal	Cable (860-001)	PLC word.bit	PLC variable (GVLMain.)	Status
-R1-003 652	Orbit 864 x6 via hub -MB01 (OT01–06)	MB01	– (none in list)	in/out 0–11	GVLIO.CommonRIO.Serial652 → Pen.Input.Orbit864Sensor[0..5]	OK
-R1-004 638 ch1	FT61 counter, cylinder 1	-XX05 1–3	PLC-PEN01-FT61-KX01 4G	12 / 13	Outlet.Cylinder[0].Input.OilFlowStatusWord / OilFlowCounter	OK
-R1-004 638 ch2	FT62 counter, cylinder 2	-XX05 5–7	PLC-PEN01-FT62-KX01 4G	14 / 15	Outlet.Cylinder[1].Input...	OK
-R1-005 638 ch1	FT63 counter, cylinder 3	-XX05 9–11	PLC-PEN01-FT63-KX01 4G	16 / 17	Outlet.Cylinder[2].Input...	OK
-R1-005 638 ch2	FT64 counter, cylinder 4	-XX05 13–15	PLC-PEN01-FT64-KX01 4G	18 / 19	Outlet.Cylinder[3].Input...	OK
-R1-007 455 ch1	QT11 CO_2 inside pen	-XX01 1/2	PLC-PEN01-QT11-KX01	20	Pen.Input.CO2	OK (no range in PLC)
-R1-007 455 ch2	PT11 total gas pressure	-XX01 3/4	PLC-PEN01-PT11-KX01	21	Pen.Input.TotalGasPressure	OK (no range in PLC)
-R1-007 455 ch3/4	spare	-XX01 5–8	–	22 / 23	–	OK
-R1-008 455 ch1	FT41 oxygen flow line 1 (symbol says FT61)	-XX01 9/10	PLC-PEN01-FT61-KX01 3G	24	OxySystem[0].Input.OxygenFlow	! drawing / cable label FT61
-R1-008 455 ch2	FT42 oxygen flow line 2	-XX01 11/12	PLC-PEN01-FT42-KX01	25	OxySystem[1].Input.OxygenFlow	OK
-R1-008 455 ch3	FT43 oxygen flow line 3	-XX01 13/14	PLC-PEN01-FT43-KX01	26	OxySystem[2].Input.OxygenFlow	OK
-R1-008 455 ch4	PT40 oxygen supply pressure	-XX01 15/16	PLC-PEN01-PT40-KX01	27	Pen.Input.OxygenInjectionPressure	! name: supply pressure, no range / alarm
-R1-009 455 ch1–4	PT61 / PT62 / PT63 / PT64 cyl. 1–2 open / close	-XX01 17–24	PLC-PEN01-PT61..64-KX01	28–31	Outlet.Cylinder[0/1].Input.PressureOpen / Close	OK (no range in PLC)
-R1-010 455 ch1–4	PT65 / PT66 / PT67 / PT68 cyl. 3–4 open / close	-XX01 25–32	PLC-PEN01-PT65..68-KX01	32–35	Outlet.Cylinder[2/3].Input.PressureOpen / Close	OK (no range in PLC)
-R1-011 455 ch1–4	spare (cable list: PT69–PT612?)	-XX01 33–40	PLC-PEN01-PT69..612-KX01	36–39	–	! cables without drawing
-R1-012 455 ch1–4	spare	-XX01 41–48	–	40–43	not in the PLC	X PLC counts 5x 750-455; read 41 words, needs 45
-R1-017 1405 ch1	spare	-XX03 1/2	–	44.0	PLC: Outlet.HydraulicPump.Input.StarterFault at 40.0	X PLC reads word 40 (= -R1-012 ch1)
-R1-017 1405 ch2–16	spare	-XX03 3–32	–	44.1–44.15	–	OK
-R1-014 563 ch1	OV41 oxygen valve line 1	-XX02 1/2	PLC-PEN01-OV41-KX01	out 20	OxySystem[0].OxyValve.Output.RawPosition	OK
-R1-014 563 ch2	OV42 oxygen valve line 2	-XX02 3/4	PLC-PEN01-OV42-KX01	out 21	OxySystem[1].OxyValve.Output.RawPosition	OK
-R1-015 563 ch1	OV43 oxygen valve line 3	-XX02 5/6	PLC-PEN01-OV43-KX01	out 22	OxySystem[2].OxyValve.Output.RawPosition	OK
-R1-015 563 ch2	spare	-XX02 7/8	–	out 23	–	OK
-R1-019 1515 ch1–8	XV61..XV64 _O / _C via relays XK01–XK08	-XX04 1–16	PLC-PEN01-XV61..64-KX01 4G	out 24.0–24.7	Outlet.Cylinder[0..3].Output.OpenValve / CloseValve	OK
-R1-020 1515 ch1	spare (rev 0: XV65 shut-off valve)	-XX04 17/18	PLC-PEN01-XV65-KX01	out 24.8	Outlet.Output.FailCloseValve ("utgår")	! cable listed, removed from drawing
-R1-020 1515 ch2	spare	-XX04 19/20	–	out 24.9	Outlet.HydraulicPump.Output.Start	X not wired on drawing
-R1-020 1515 ch3	spare	-XX04 21/22	–	out 24.10	Outlet.HydraulicPump.Output.Reset	X not wired on drawing
-R1-020 1515 ch4–8	spare	-XX04 23–32	–	out 24.11–15	–	OK

Not on the drawings, used or expected by the PLC / FD: hydraulic tank fault LSL61/TSH61 (Outlet.Input.HydraulicOilTankFault, not mapped); emergency stop (GVLMain.EmergencyStop, OxySystem[i].Input.EmergencyStop, never written); CommonRIO field power / fuse status (Status.FieldPowerOK / FieldFuseOK default TRUE); hydraulic pump starter fault and start / reset (above).

Process image check (750-362, complex and analog modules first, digital packed after). Drawing rev 1: inputs 0–11 (652), 12–19 (2x 638), 20–43 (6x 455), 44 (1405) = 45 words; outputs 0–11, 12–19, 20–23 (2x 563), 24 (2x 1515) = 25 words. PLC: read 41 words with DI at word 40, write 25 words. Either remove one 750-455 (-R1-012) or change the PLC to read 45 words with the DI at word 44.

Title **I/O list 1/2: CommonRIO -R1-001 – drawing vs PLC**	Drawing no. **CAS-PID-900**	
System I/O list, PLC/RIO1 cabinet (792.001 rev 1) vs FB.RIO.CommonRIO	Project 15169 CAS Heimdall ScaleAQ	Rev. / date A / 2026-10-16

Module / ch	Drawing: tag, description	Terminal	Cable (860-001)	PLC word.bit	PLC variable (GVLMain.)	Status
-R1-106 455 ch1	LT0n level inside pen	-XX06 1/2	RIOn-PEN01-LT0n-KX01	0	Inlet[i].Input.LevelInsidePen	OK
-R1-106 455 ch2	LT2n level outside pen	-XX06 3/4	RIOn-PEN01-LT2n-KX01	1	Inlet[i].Input.LevelOutsidePen	OK
-R1-106 455 ch3	PT4n pressure after oxy pump (PT41/42/43)	-XX06 5/6	RIOn-PEN01-PT4n-KX01	2	OxySystem[i].Input.OxyPumpPressure	OK
-R1-106 455 ch4	PT(n+3) pressure after injector (PT44/45/46)	-XX06 7/8	RIOn-PEN01-PT4(n+3)-KX01	3	OxySystem[i].Input.OxynatorOutletPressure	OK (FD numbers differ)
-R1-101 1405 ch1	XF05 24 V field breaker tripped	internal	–	4.0	Inlet[i].Input.FieldPower24VTripped	OK
-R1-101 1405 ch2	XF05 load $>$ 80 %	internal	–	4.1	Inlet[i].Input.FieldPower24VHighLoad	OK (no alarm in PLC)
-R1-101 1405 ch3	XF01 230 V breaker tripped	internal	–	4.2	Inlet[i].Input.MainCircuitBreakerTripped	OK (no alarm in PLC)
-R1-101 1405 ch4	UPS XT01 backup power on	internal	–	4.3	Inlet[i].Input.BackupPowerOn	! RIO2/3: wired to +24 V, always TRUE
-R1-101 1405 ch5	UPS XT01 battery low / faulty	internal	–	4.4	Inlet[i].Input.BatteryFault	! RIO2/3: wired to +24 V, always TRUE
-R1-101 1405 ch6–16	spare	–	–	4.5–4.15	–	OK
-R1-104 1405 ch1	LS7n1H water detected, chamber 1	-XX07 1/2	RIOn-PEN01-LS7n1H-KX01	5.0	Inlet[i].Input.BilgePumpSwitchHigh[0]	OK (no alarm in PLC)
-R1-104 1405 ch2	LS7n1HH high-high, chamber 1	-XX07 3/4	RIOn-PEN01-LS7n1HH-KX01	5.1	Inlet[i].Input.BilgePumpSwitchHighHigh[0]	! no alarm, no 2oo3 trip
-R1-104 1405 ch3 / ch4	LS7n2H / LS7n2HH, chamber 2	-XX07 5–8	RIOn-PEN01-LS7n2H/HH-KX01	5.2 / 5.3	...SwitchHigh[1] / HighHigh[1]	OK
-R1-104 1405 ch5 / ch6	LS7n3H / LS7n3HH, chamber 3	-XX07 9–12	RIOn-PEN01-LS7n3H/HH-KX01	5.4 / 5.5	...SwitchHigh[2] / HighHigh[2]	OK
-R1-104 1405 ch7	QE7n bilge pump breakers	-XX07 13/14	RIOn-PEN01-QE7n-KX01	5.6	Inlet[i].Input.CircuitBreakerBilgePump	OK
-R1-104 1405 ch8	PU41-QS oxy pump leakage (RIO1 only)	-XX07 15/16	DC1-RIO1-KX01	5.7	OxySystem[i].Input.OxyPumpLeakDetedtion	! RIO2/3 spare; not used in logic
-R1-104 1405 ch9	PU41-TS oxy pump temperature (RIO1 only)	-XX07 17/18	DC1-RIO1-KX01	5.8	OxySystem[i].Input.OxyPumpHighTemperature	! RIO2/3 spare; not used in logic
-R1-104 1405 ch10–16	spare	-XX07 19–32	–	5.9–5.15	–	OK
-R1-102 1515 ch1	reset XF05	internal	–	out 0.0	Inlet[i].Output.ResetFieldPower24V	OK (no logic writes it)
-R1-102 1515 ch2–8	spare	–	–	out 0.1–0.7	–	! DC n expects relays XK1–3 for bilge contactors
ATV650 DC n	oxy pump drive, embedded Ethernet	Cat6	NC-DCn-KY01	8501/8502, 3201–3211, 7121	OxySystem[i].OxyPump (InletRIO[i] DriveMaster)	! Ethernet port not drawn on 874.004

Process image check. Drawing: inputs 1 x 455 (words 0–3) + 2 x 1405 (words 4, 5) = 6 words; outputs 1 x 1515 = 1 word (8 bits). PLC: FC23 read 26 words / write 21 words from address 0. Check on the real coupler that reading and writing beyond its process image is accepted (the simulator serves 64 words).

Inlet n: n = 1 InletRIO1 = -R1-100 in the PLC/RIO1 cabinet (792.001 sh.26–31); n = 2 RIO2 cabinet (792.002, one generic drawing); n = 3 RIO3 cabinet (no drawing; from the cable list). PLC index i = n − 1.

Title **I/O list 2/2: InletRIO 1–3 -R1-100 – drawing vs PLC**	Drawing no. **CAS-PID-901**	
System I/O list, PLC/RIO1 and RIO2/3 cabinets vs FB.RIO.InletRIO	Project 15169 CAS Heimdall ScaleAQ	Rev. / date A / 2026-10-16

No.	Cl.	Finding	Sources	Proposed action
A1	A	CommonRIO has six 750-455 on drawing 792.001 rev 1 (-R1-007..012); the PLC reads 41 words with five, so its DI word 40 is really -R1-012 ch1 and the DI module (word 44) is not read.	792.001 sh.07, 13–18; FB.RIO.CommonRIO	Decide: remove -R1-012, or read 45 words with DI at 44 (CAS-PID-900).
A2	A	Hydraulic pump start / reset (DO 24.9 / 24.10) and starter fault (DI 40.0) are on channels the drawing shows as spare.	792.001 sh.22, 25; FB.RIO.CommonRIO	Wire them, or move them to the channels the drawing assigns.
A3	A	Emergency stop is not wired to any input; GVLMain.EmergencyStop and OxySystem[i].Input.EmergencyStop are never written. FD / action list: Framo E-stop shall stop PU4n and close OV4n.	AL 16–17; FD 3.2.4; PLC	Use Framo_n.EmergencyStop (OPC UA) per line, or a hardwired loop (Framo XSTO / X2).
A4	A	Orbit 864 EnableForControl defaults to FALSE and is not retained: after a restart OIC01 has no PV until set online.	PLC Orbit864ItemConfigDUT; ConfigValues	Retain it (Orbit864SensorConfig) with OT01–03 = TRUE.
A5	A	No 2oo3 LS7nmHH trip of the inlet pump and no bilge alarms in the PLC (inputs are read).	FD 3.2.6; 792.001 sh.29	Add AlarmManagement + interlock I2.
A6	A	Bilge pump contactors in DC n are switched by "RIO relays -XK1/2/3" that no RIO drawing has, and the PLC has no outputs.	874.004 sh.05; 792.001/002	Clarify: self-controlled pumps (FD) or PLC-controlled (action list 6/47/77).
A7	A	RIO2 / RIO3: UPS inputs 4.3 / 4.4 wired to +24 V (no UPS) read as BackupPowerOn / BatteryFault = TRUE.	792.002 sh.07; FB.RIO.InletRIO	Ignore them for RIO2/3 in the PLC, or rewire as spare.
A8	A	PLC InletRIO reads 26 / writes 21 words; the coupler image is 6 in / 1 out.	FB.RIO.InletRIO; 792.001/002	Check on a real 750-362; reduce to 6 / 1 if refused.
A9	A	Ranges missing in the PLC (DUT default 0–10): QT11, PT11, PT40, PT61–68.	PLC ConfigValues	Add ranges from the datasheets (PT61–68: 0–250 bar).
B1	B	Oxygen-branch pressure transmitters: FD PT41/43/45 in, PT42/44/46 out; drawings / cable list PT41–43 after the pump, PT44–46 after the injector.	FD 3.2.4; 792.001 sh.31; 792.002 sh.10	Drawn as the cabinet drawings; update the FD.
B2	B	Orbit 864 tags: FD inside OT01–03, outside OT21–23; drawing OT01–OT06 on hub ports 1–6.	FD table 17; 792.001 sh.11	Choose one set; record which slave is where.
B3	B	LIC01: FD has P scheduled on pump count (0.5/1/2), I 600 s, 0–20 cm scaling, L/H and deviation alarms; PLC has one gain set (Kp 450, Ti 60 s) and no level alarms.	FD table 8; ConfigValues	Decide; add alarms.
B4	B	OIC01: FD outputs 0–100 % to OV41–43 (or cascade to FIC), PV OT03 with selector; PLC outputs total kg/h shared between lines, PV = mean of EnableForControl sensors.	FD tables 13–14; FB.Inlet.InletSupervisor	Update the FD to the implemented structure, or change the PLC.
B5	B	FD alarm "low differential pressure over the injector" (PT44–PT41) is not in the PLC.	FD table 15	Add.
B6	B	PT40 oxygen supply pressure: named OxygenInjectionPressure in the PLC; no low alarm (FD / action list: oxygen not available).	792.001 sh.14; AL 14/22/38	Rename, add PAL40.
B7	B	Inlet group start only with the outlet open (PLC function flow diagram) is not in the PLC.	PLC flow diagram p.3	Decide.
B8	B	Oxy pump leakage / temperature DIs exist for line 1 only (RIO1); DC2/DC3 send them, RIO2/3 have no inputs; the PLC does not use them.	874.004 sh.05; 792.001/002	Add inputs on RIO2/3 and alarms.
B9	B	Framo pump: cabinet drawing 874.001 shows SX1000; FD and the folder have SR2000.	874.001; FD 3.2.1	Confirm with Framo.
B10	B	Hydraulics: FD has PT61–68 on both sides, XV65 and an accumulator; newest Servi schematic A1 has 4 transmitters, no XV65. Drawing R1 has 8 PTs, XV65 removed; cable list still has XV65 and PT69–PT612.	FD 3.2.3; Servi A1; 792.001; 860-001	Align FD, Servi and cable list.
B11	B	FT41–43 range / unit: PLC 0–150 kg/h, FD FIC scaling "0–20" (unit unclear); flowmeter make Linde mass flow vs Normex VA500 quote.	FD table 14; Linde; ConfigValues	Confirm the transmitter range.
B12	B	Framo heartbeat: Framo spec HOLD → REMOTE automatically when the heartbeat returns; FD (and simulator) require a Reset.	Framo spec rev B; FD table 5	Confirm with Framo.
C1	C	FT41 field symbol and cable labelled FT61; cable number PLC-PEN01-FT61-KX01 used twice.	792.001 sh.14; 860-001 r.66/90	Correct drawing and cable list.
C2	C	QT11 (CO_2) drawn with an "O2 transmitter" symbol.	792.001 sh.13	Correct the symbol.
C3	C	No network drawing: switch, IP plan, ATV650 Ethernet ports and the PLC-cabinet internal links are not drawn; one Cat6 to the PLC/RIO1 cabinet.	792.001; 874.004; 860-001	Make a network sheet with IP plan.
C4	C	RIO3 has no drawing; RIO2/3 drawing is generic with copied RIO1 tags (LS711..QE71).	792.002	Issue RIO2 and RIO3 sheets.
C5	C	Orbit 864 cables and their 24 V supply, and the Linde cabinet terminals, are not on the CAS drawings or in the cable list.	792.001 sh.11; 860-001	Add.
C6	C	Cable list copy errors: DC2/DC3 cables prefixed DC1-, bilge pumps 3.x tagged OP31–33, template cover sheet.	860-001	Correct.
C7	C	Drive cabinet DC1 supply cable 5G2.5 vs 5G6 for DC2/DC3; 25 A upstream for a 15 kW ATV650 plus bilge pumps.	875.002; 874.004	Check the cable sizing.
C8	C	The FD's own P&ID reference table (1.1) and PLC–RIO fail-safe table (2.2) are empty.	FD rev B	Fill in, referring to this drawing set.

Title: **Open points**		Drawing no.: **CAS-PID-950**
System: All systems	Project: 15169 CAS Heimdall ScaleAQ	Rev. / date: A / 2026-10-16